DRV8825-Based Bipolar Driver with Optical Isolation and DIP-Selectable Microstepping

1 Features

- DRV8825-based bipolar stepper motor driver
- Wide motor supply voltage range: 12 V to 24 V
- Output current up to 2.5 A per phase (peak), adjustable via on-board precision trimmer
- Optically isolated STEP and DIR control inputs for improved noise immunity and system protection
- Standard STEP/DIR interface, compatible with microcontrollers, CNC controllers and PLCs
- DIP switch–selectable microstepping modes (Full, 1/2, 1/4, 1/8, 1/16, 1/32 step)
- Integrated dual H-bridge with current regulation and mixed decay control
- On-board 5 V linear regulator for logic, opto-isolators and driver core
- Dedicated current sense resistors for accurate phase current control
- Integrated fault reporting and protection features (overcurrent, thermal shutdown, undervoltage lockout)
- Power status LED indicator
- Screw terminal connectors for motor phases and power supply

2 Applications

- CNC machines and motion control systems
- 3D printers and custom additive manufacturing platforms
- Industrial automation and positioning systems
- Robotics and mechatronic control applications
- PLC-controlled motion systems
- Linear actuators and precision positioning stages
- Laboratory and test bench motor control setups
- Educational platforms for motor control and power electronics
- Custom motion controllers for embedded systems

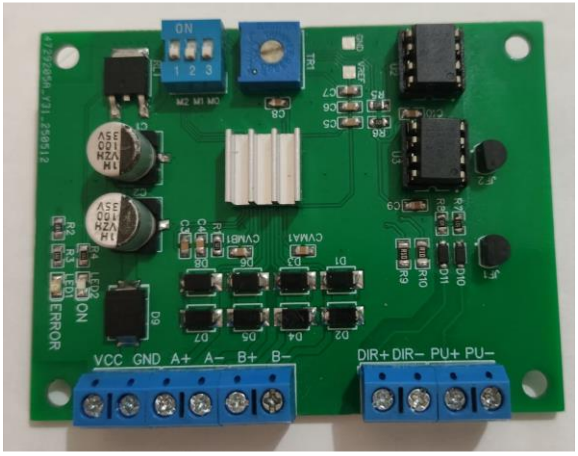

Figure 1. Prototype

3 Description

The Isolated Bipolar Stepper Motor Driver Module is a compact and robust motion control solution designed to drive bipolar stepper motors in professional prototyping, automation and embedded control applications. The module is based on the DRV8825 stepper motor driver IC, which integrates dual H-bridge power stages, fixed off-time current regulation and advanced microstepping capability within a single device.

The driver operates from a 12 V to 24 V motor supply and is capable of delivering up to 2.5 A peak output current per phase, with continuous current levels typically up to 2.0 A per phase, depending on thermal conditions and airflow. The maximum phase current is set through an on-board precision trimmer, while dedicated current sense resistors provide accurate and stable current regulation over the full operating range.

Motor control is performed through a standard STEP/DIR interface, with both control inputs featuring optical isolation implemented using high-speed opto-isolators. This isolation significantly improves noise immunity and protects upstream control electronics—such as microcontrollers, CNC controllers and PLCs—

from ground loops, voltage transients and electromagnetic interference commonly present in motor drive systems.

Microstepping resolution is selectable via an on-board DIP switch, allowing configuration from full-step operation up to 1/32 microstepping. This flexibility enables smoother motor motion, reduced mechanical resonance and improved positioning resolution without requiring firmware changes in the host controller.

An integrated 5 V linear regulator supplies the driver logic, opto-isolators and auxiliary circuitry, eliminating the need for an external logic power supply and simplifying system integration. Separate power domains and localized bulk and ceramic decoupling capacitors are employed to minimize noise coupling between the motor power stage and the control circuitry.

The module incorporates the comprehensive protection features of the DRV8825, including overcurrent protection, thermal shutdown and undervoltage lockout, contributing to reliable and safe operation under fault or overload conditions. Screw terminal connectors are used for motor phases and power input, providing a secure and practical interface for embedded, industrial and laboratory environments.

This module is intended for professional prototyping, motion control development and educational demonstration of power electronics and stepper motor control principles. It is not designed or certified for use in safety-critical, automotive or medical applications.

Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Motor supply voltage (VM)		12	-	24	V
Logic supply voltage	On-board regulator	-	5.0	-	V
Peak output current per phase	Short duration, adequate cooling	-	-	2.5	A
Continuous output current per phase	Dependent on thermal conditions	-	-	2.0	A
Microstepping resolution	DIP switch selectable	Full	-	1/32	
Control input voltage (PU, DIR)	PCB input, current-limited	5	-	24	V
STEP input frequency	Opto-isolated interface	-	-	250	kHz
Control input isolation voltage (6N137)	Opto-isolators (6N137)	-	-	2.5	kVrms
Operating temperature range		-20	-	85	°C
Storage temperature range		-40	-	125	°C

Notes:

- Peak output current values must not be sustained continuously and depend on thermal conditions, PCB copper area and airflow.
- Continuous output current capability is typically up to 2.0 A per phase under adequate cooling.
- Phase current is set via the on-board trimmer according to the DRV8825 VREF equation.
- PU (STEP) and DIR inputs are current-driven and optically isolated; input current is internally limited by a JFET-based protection stage.
- Input current may vary due to JFET IDSS tolerance.
- Control input voltages above 24 V or severe transients may exceed component ratings unless additional external protection is provided.

BOM (Bill of Materials)

Name	Designator	Quantity
100uF 35V	C1,C2	2
10nF	C3	1
0.1u	C4,C5,C6,C8,C9,C10,CVMA1,CVMB1	8
0.27u	C7	1
SS54	D1,D2,D3,D4,D5,D6,D7,D8	8
SK510C	D9	1
1N4148W	D10,D11	2
2N5485	JF1,JF2	2
LED-0805_R	LED1	1
LED-0805_G	LED2	1
CONN-TH_2P-P5.00	P1,P2,P3,P4,P5	5
1M	R1	1
10k	R2	1
1k	R3,R4	2
4k7	R5,R6	2
330R	R7,R8	2
0R10	R9,R10	2
KIA7805AF-RTF/P	RL1	1
KF1001-03P	SW1	1
3386F-10K	TR1	1
DRV8825PWPR	U1	1
HT-6N137	U2,U3	2

PCB size

60.83mm X 80.77mm

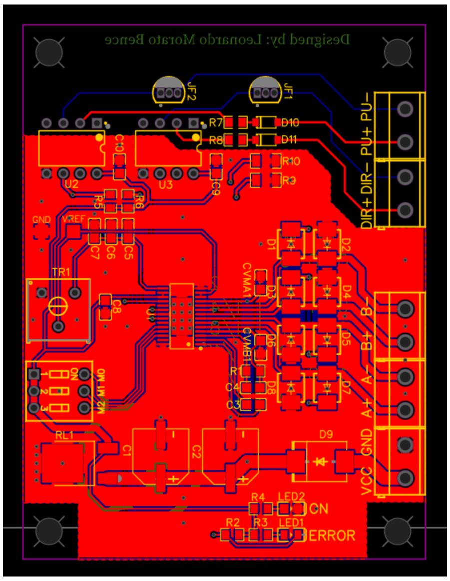

Figure 2. Top Layer

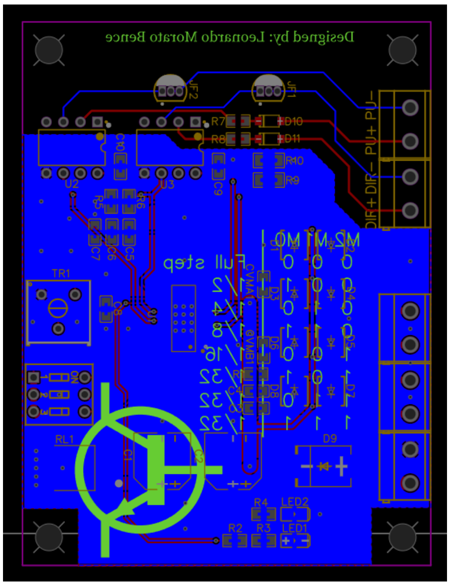

Figure 3. Bottom Layer

Block Diagram

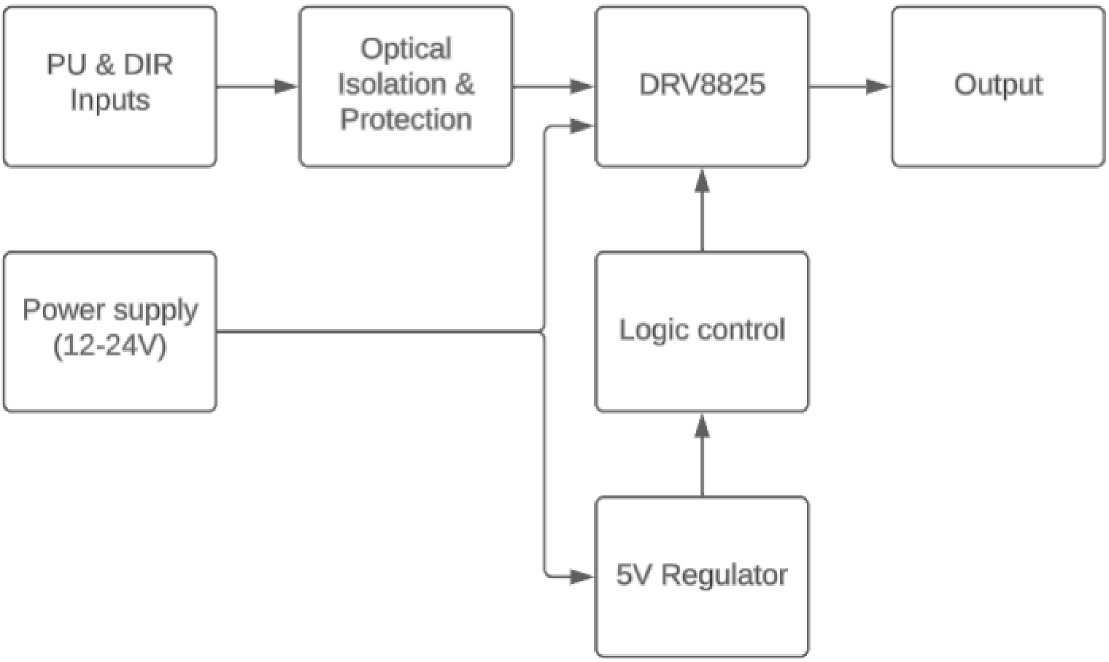

Figure 4. Block Diagram

Block Diagram Description

Figure 4 illustrates the functional block diagram of the **Isolated Bipolar Stepper Motor Driver Module**.

PU (STEP) and DIR control signals enter the module through a dedicated **optical isolation and protection stage**, which provides current limiting, transient protection and galvanic isolation from the external controller. This input stage allows reliable operation with control voltages in the **5 V to 24 V** range while improving noise immunity and protecting upstream electronics.

The isolated control signals are then processed by the **DRV8825 stepper motor driver**, which performs step decoding, direction control, microstepping generation and current regulation. Motor phase outputs are generated by the integrated dual H-bridge power stage and are routed directly to the motor output terminals.

The motor power stage is supplied directly from the external **12 V to 24 V power supply**, while an on-board **5 V linear regulator** generates a regulated logic supply for the isolation circuitry and configuration logic. The **logic control block** includes hardware configuration elements such as the microstepping selection (DIP switch), enable and reset control, and the adjustable current reference circuitry used to set the motor phase current.

By separating control, logic and power domains, the architecture ensures robust operation in electrically noisy environments and simplifies integration into industrial, automation and embedded motion control systems.

Schematic

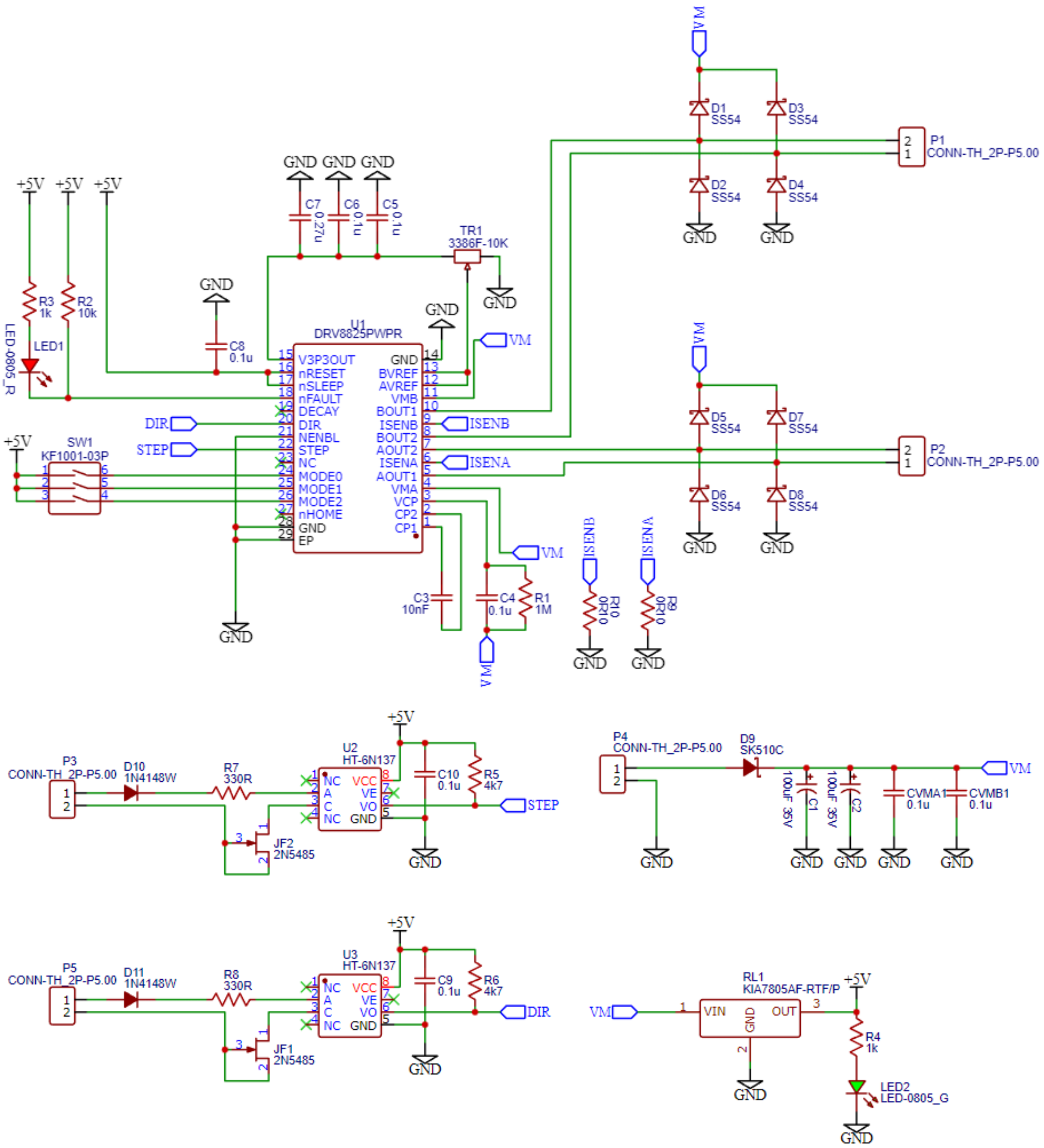

Figure 5. Schematic

3D image of the PCB

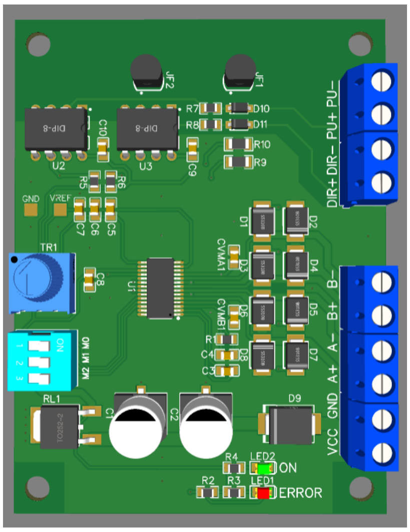

Figure 6. PCB Top Side

Figure 7. PCB Bottom Side

Disclaimer

The information contained in this document is provided “as is” and is subject to change without notice. This product is intended solely for educational, prototyping, and demonstration purposes. It is not certified for use in safety-critical, medical, automotive, or industrial applications. The author assumes no responsibility for any damages, direct or indirect, resulting from the improper or incorrect use of this module.

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